PENGZHEN LI（李鹏振）

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EDUCATION

University of Tennessee, Knoxville expected 2027

Ph.D. in Economics — Committee: James Lake, Georg Schaur, Matthew Van Essen [[confirm]]

Shandong University 2018

M.A. in Finance — thesis published in *Technological Forecasting and Social Change* (950+ citations)

Tongji University, Shanghai 2015

B.A. in Economics

RESEARCH INTERESTS

International Trade, Environmental and Energy Economics, Applied Econometrics, and Political Economy.

PEER-REVIEWED PUBLICATIONS

Four SSCI/SCI-indexed publications.

"Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data." With K. Du and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303. **950+ citations (Google Scholar).**

"Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability." With T. Yu et al. *Biomass and Bioenergy*, 208 (2026), 108793.

"Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA." With T. Yu et al. *Transportation Research Record*, 2678(9) (2024), 639–654.

"Beef price spread relationship with processing capacity utilization." With C. Martinez et al. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

WORKING PAPERS

Trade Restrictiveness Indices with Discriminatory Tariffs with James Lake [JMP] — application: US–China tariff war, 2018–2024

Abstract: Trade-restrictiveness indices summarize a country's tariff schedule as a single welfare-equivalent uniform tariff, but the existing indices assume most-favored-nation tariffs. Since 2018 the largest tariff changes—the US–China tariff war, retaliatory tariffs, and proliferating preferential agreements—have been discriminatory by design. We extend the trade-restrictiveness index to partner-specific tariffs while preserving its welfare interpretation and derive a closed-form decomposition into a "level" component (how much a country restricts trade) and a "discrimination" component (how unevenly it does so). Computing the indices from WITS/Comtrade data for 2017–2024, we find that the MFN-only index systematically understates trade distortions when discrimination is high, and the decomposition separates, for the first time in this framework, the welfare cost of discrimination from the cost of protection. [[confirm / replace with paper abstract]]

The Impact of Mayoral Endorsements on Presidential Election Outcomes with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).*

Abstract:

Welfare Analysis of a Forest-Residue Sustainable Aviation Fuel Supply Chain under Endogenous Demand and Price Uncertainty with T. Edward Yu

Stochastic Optimisation of Logging Residue–based Biofuel Supply Chains Considering Feedstock Ash Content and Biofuel Price with T. Yu et al.

Sustainable Aviation Fuel Production from Forest Residues with Ash Content: A Case Study of Nashville Airport with T. Yu et al.

WORK IN PROGRESS

Cooperative Multilateral Tariff Negotiations: A Quantitative Analysis Using Shapley Value with Georg Schaur — RCEP / CPTPP scenario analysis

PRESENTATIONS

Midwest Economic Theory and International Trade Meetings, Virginia Tech — Spring 2025
Western Regional Science Association, Annual Meeting, Hawaii — 2023
Western Agricultural Economics Association, Annual Meeting, Santa Fe, NM — 2022
Southern Agricultural Economics Association, Annual Meeting, New Orleans, LA — 2022

PROFESSIONAL EXPERIENCE

University of Tennessee – Knoxville, TN — 2020 – present
Graduate Research & Teaching Assistant. USDA NIFA and FAA ASCENT projects (2020–2023): mixed-integer supply-chain optimization (GAMS), IMPLAN/DEA siting analysis, and spatial data pipelines (ArcGIS, R, Python) for forest-residue sustainable aviation fuel; two publications and three working papers.

Shandong University – Jinan, China — 2016 – 2018
Graduate Research Assistant. Panel-threshold and machine-learning analysis of green innovation and CO_2 emissions (published in *TFSC*, 2019).

TEACHING EXPERIENCE

University of Tennessee

Instructor

ECON 351: Monetary Economics	Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development	Summer 2025
ECON 313: Intermediate Macroeconomics	Summer 2026

Teaching Assistant

Ph.D. courses:

ECON 511: Microeconomic Theory	Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics	Spring 2024

Undergraduate and master's courses:

ECON 581: Mathematical Methods in Economics	Summer 2024
AREC 525: Agribusiness Operations Research Methods	Spring 2022, Spring 2023

HONORS AND AWARDS

J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
Graduate Assistantship, University of Tennessee	2020 – present
Outstanding Volunteer, migrant-family tutoring program, Shanghai	[[year]]

SKILLS

Programming Languages: Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS, IMPLAN.
Methods: panel and causal econometrics (FE, IV, DiD, event study, SCM, GMM, panel threshold), time series (TVAR, VECM, mGARCH), mixed-integer and stochastic programming, DEA, cooperative and mean-field games, machine learning.
Credentials: CFA Level II.
Languages: Chinese (native), English (fluent).

REFERENCES

James Lake
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Department of Economics
University of Tennessee
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Georg Schaur
Leigh Burch III Economics Professor and Haslam Family Faculty Fellow
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T. Edward Yu
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